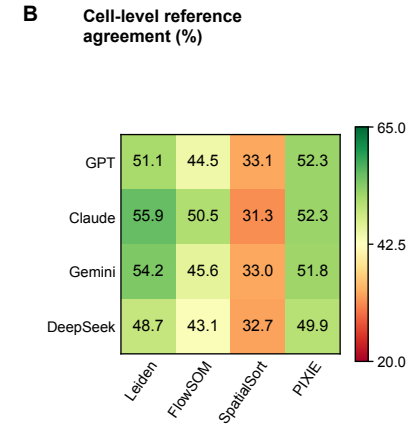
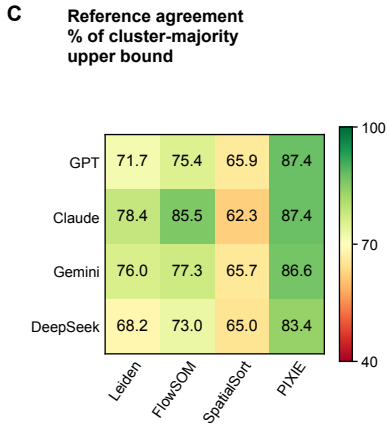
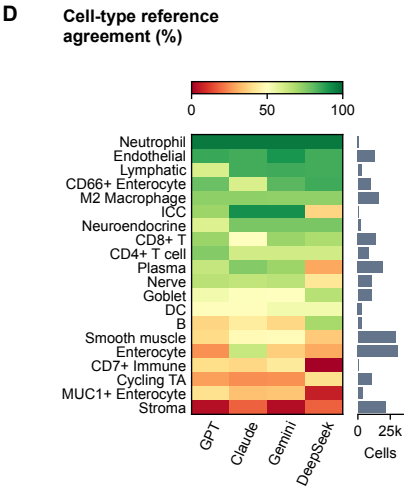
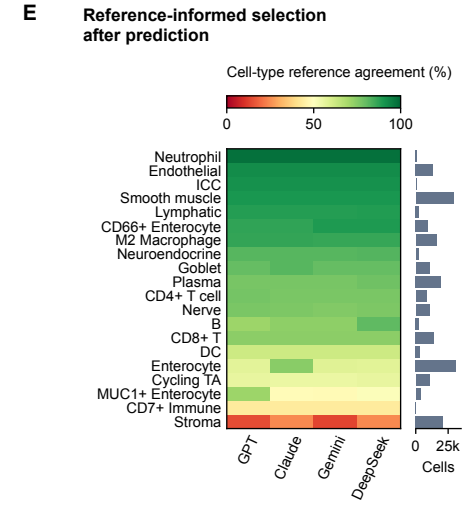
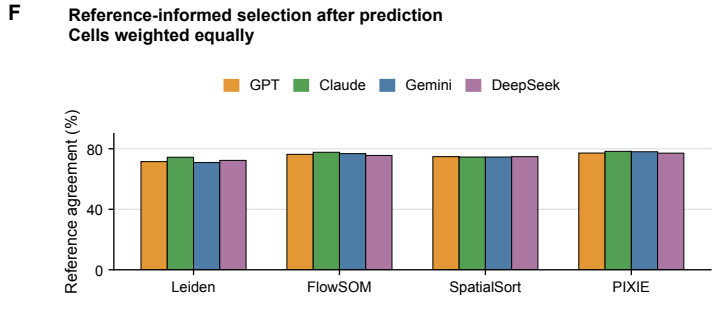

A
CODEX Features
Clustering
Cluster Features
Optimization
Annotation
Evaluation
Cell 1, Cell 2, Cell 3
x, y, Marker 1,
Marker 2, ...
Leiden
FlowSOM
SpatialSort
PIXIE
{
"cluster 1":
["marker 1", "marker 2", ...],
"cluster 2":
["marker 1", "marker 2", ...]
}
1. Expression Threshold
2. Fraction Threshold
3. Mean Expression Threshold
4. Maximum Number of Markers
5. Minimum Number of Markers
Gemini
GPT
DeepSeek
Claude
Reference
comparison

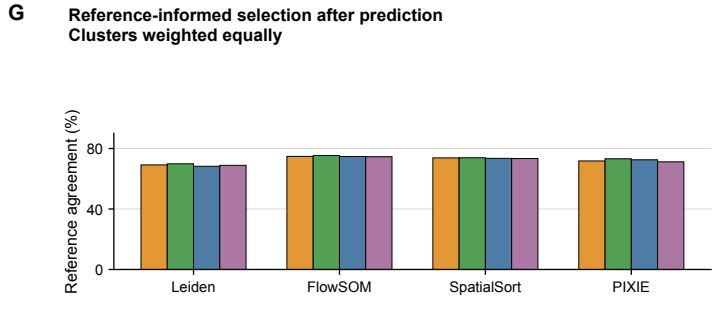

B
Cell-level reference
agreement (%)
GPT 51.1 44.5 33.1 52.3
Claude 55.9 50.5 31.3 52.3
Gemini 54.2 45.6 33.0 51.8
DeepSeek 48.7 43.1 32.7 49.9
65.0
42.5
20.0
Leiden
FlowSOM
SpatialSort
PIXIE

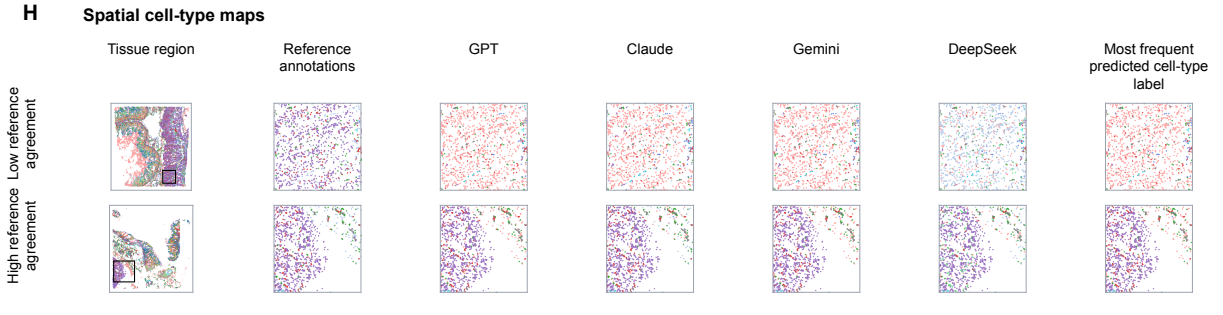

C
Reference agreement
% of cluster-majority
upper bound
GPT 71.7 75.4 65.9 87.4
Claude 78.4 85.5 62.3 87.4
Gemini 76.0 77.3 65.7 86.6
DeepSeek 68.2 73.0 65.0 83.4
100
70
40
Leiden
FlowSOM
SpatialSort
PIXIE


D
Cell-type reference
agreement (%)
0
50
100
Neutrophil
Endothelial
Lymphatic
CD66+ Enterocyte
M2 Macrophage
ICC
Neuroendocrine
CD8+ T
CD4+ T cell
Plasma
Nerve
Goblet
DC
B
Smooth muscle
Enterocyte
CD7+ Immune
Cycling TA
MUC1+ Enterocyte
Stroma
GPT
Claude
Gemini
DeepSeek
0
25k
Cells


E
Reference-informed selection
after prediction
Cell-type reference agreement (%)
0
50
100
Neutrophil
Endothelial
ICC
Smooth muscle
Lymphatic
CD66+ Enterocyte
M2 Macrophage
Neuroendocrine
Goblet
Plasma
CD4+ T cell
Nerve
B
CD8+ T
DC
Enterocyte
Cycling TA
MUC1+ Enterocyte
CD7+ Immune
Stroma
GPT
Claude
Gemini
DeepSeek
0
25k
Cells


F
Reference-informed selection after prediction
Cells weighted equally
GPT
Claude
Gemini
DeepSeek
Reference agreement (%)
80
40
0
Leiden
FlowSOM
SpatialSort
PIXIE


G
Reference-informed selection after prediction
Clusters weighted equally
Reference agreement (%)
80
40
0
Leiden
FlowSOM
SpatialSort
PIXIE


H
Spatial cell-type maps
Tissue region
Reference annotations
GPT
Claude
Gemini
DeepSeek
Most frequent predicted cell-type label
Low reference agreement
High reference agreement